Vitor Alexandre Garcia Vaz

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Summary

Computer Engineering student at USP São Carlos. Experience building REST APIs, event-driven services, real-time data pipelines and web interfaces, with code running in production in an online store with PIX payments and in telemetry and IoT platforms. Hands-on with layered architecture, containerization and technical documentation.

Education

University of São Paulo (USP) — EESC and ICMC São Carlos, SP
Bachelor's in Computer Engineering *2023 - 2027 (expected)*

Experience

Undergraduate Research LaSDPC — ICMC-USP 04/2025 - Present
- Developed a Java/Spring Boot platform that ingests, processes and stores IoT device data in real time, solving the problem of scattered sensor data with no queryable history for the team
- Implemented the Spring Boot REST API that exposes this data and allows sending commands to devices remotely, with its own authentication and health monitoring already in production
- Resolved the mismatch between the Network Server's and the application's device identities by writing a Python bridge that translates and validates messages between the two systems

Avionics Coordinator TOPUS Aerospace Projects — EESC-USP 05/2025 - 09/2026
- Developed the C++/FreeRTOS embedded firmware used in 5 of the group's rocket launches, covering sensor acquisition, flight state machine and critical system deployment
- Standardized radio communication and critical data persistence into reusable C++ libraries, eliminating code duplication across the group's projects

Project Manager SAEComp — USP 05/2025 - 03/2026
- Gathered requirements with the department office, designed the PostgreSQL database and led the team that built the Node.js/TypeScript online store, from data modeling to the payment interface; the store increased the office's profit 75-fold
- Integrated PIX payments with automatic real-time confirmation, including automatic order cancellation when a charge fails, eliminating the manual reconciliation that existed before
- Structured the Express application to be easy to maintain and extend, with data validation, centralized error handling and always up-to-date Swagger API documentation

Embedded Systems Department Director Zenith Aerospace — EESC-USP 03/2024 - 06/2025
- As Director of the Embedded Systems Department, trained new members and managed the subsystem
- Resolved C driver duplication across the group's platforms, refactoring a reusable C abstraction layer across different microcontrollers (such as STM32, Arduino and ESP32)

Projects

IoT Monitoring Platform for Coffee Farming | *Java, Spring Boot, Kafka, PostgreSQL, Docker* 06/2026 - 09/2026
- MQTT → Kafka → Postgres time-series ingestion and a Spring Boot REST API with JWT and Prometheus

LPWAN ↔ Application Bridge | *Python, Pydantic, Redis, Prometheus* 05/2026 - 09/2026
- Bidirectional Python bridge between the Network Server and the application, mapping DevEUI ↔ MAC in Redis

SAEComp Store Backend | *Node.js, TypeScript, Express, PostgreSQL, Zod* 08/2025 - 02/2026
- Cart, orders and inventory control, with a rating and review system for users
- Express admin dashboard with image upload and API documentation via Swagger

Technical Skills

Languages: TypeScript, JavaScript, Python, Java, C++, C, SQL
Backend: Node.js, Express, FastAPI, Spring Boot, REST, WebSocket, SSE, JWT, Zod, Pydantic
Data & Messaging: PostgreSQL, TimescaleDB, Redis, Apache Kafka, Kafka Connect, MQTT
Front-end: React, Vite, Recharts, three.js
DevOps & Tools: Docker, Docker Compose, Prometheus, Git, Linux, Swagger, LaTeX
Spoken Languages: Portuguese (native), English (advanced)

Awards

Best Paper Award, IEEE SmartIoT 2026 — co-author of a paper on a reproducible, observable IoT testbed
2nd place — CubeSat Category, Latin American Space Challenge (LASC) 2026 — Atena CubeSat (TOPUS)